

BRAC UNIVERSITY
CSE331 : Automata and Computability
Summer 2026

Duration: 30 minutes

Quiz 2

Total: 22 marks

Name:

ID:

Section:

Q1) (12 points) Let $\Sigma = \{0, 1\}$. Give regular expressions generating each of the following languages over Σ .

$L_1 \rightarrow \{w \in \Sigma^* \mid w \text{ contains 1 in the second position}\}, \text{ where } \Sigma = \{0, 1\}$

$L_2 \rightarrow \{w \in \Sigma^* \mid |w| \% 3 \neq 1\}, \text{ where } \Sigma = \{0, 1\}$

$L_3 \rightarrow \{w \in \Sigma^* \mid \text{Every even position of } w \text{ is } 1\}, \text{ where } \Sigma = \{0, 1\}$

$$L_4 = L_1 \cap \overline{L_2}$$

(a) Write a Regular Expression for L_1 (2 points)

$\Sigma 1 \Sigma^*$

(b) Write a Regular Expression for $\overline{L_1}$ (2 points)

$\epsilon + \Sigma + \Sigma 0 \Sigma^*$

(c) Write a Regular Expression for L_2 . (2 Points)

$(\Sigma \Sigma \Sigma)^* (\epsilon + \Sigma \Sigma)$

(d) Write a Regular Expression for $(\overline{L_2})^*$. (1 Point)

$(\Sigma \Sigma \Sigma)^* \Sigma$

(e) Write all four letter strings in L_4 . (2 Points)

0100, 0101, 0110, 0111, 1100, 1101, 1110, 1111

(f) Write a Regular Expression for L_4 . (2 Points)

$\Sigma 1 \Sigma \Sigma (\Sigma \Sigma \Sigma)^*$

(g) Is $L_3 \subset L_1$? (1 Point)

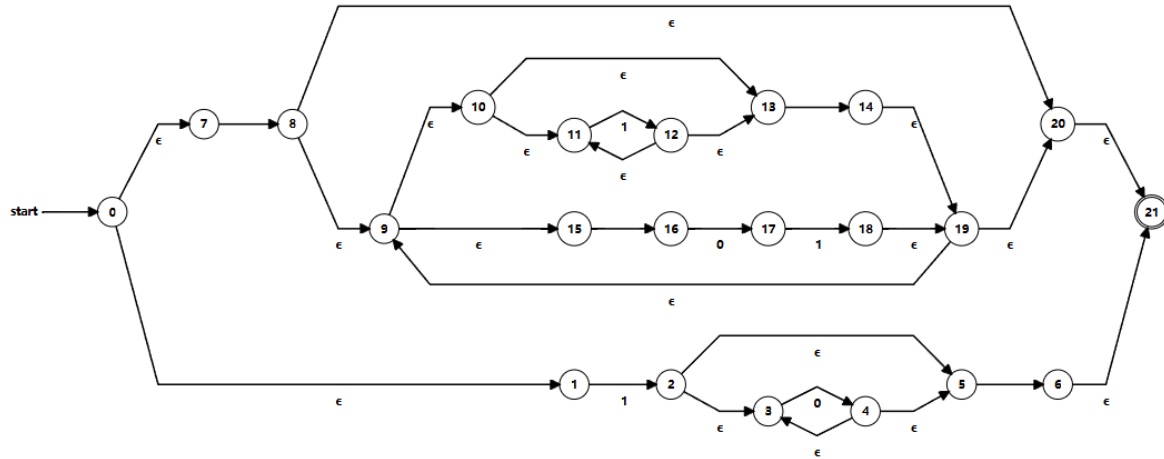
No

$0 \in L_1$

$0 \notin L_3$

Q2) (8 points) Convert the following RE into an equivalent NFA using the Thompson's construction

$10^* \cup (1^* \cup 01)^*$



Bonus

$L_1 \rightarrow \{w \in \Sigma^* \mid w \text{ contains } 1 \text{ in the second position}\}$, where $\Sigma = \{0, 1\}$

$L_2 \rightarrow \{w \in \Sigma^* \mid |w| = 3\}$, where $\Sigma = \{0, 1\}$

Write a Regular Expression for the following regular language: (2 Points)

$$L(M) \rightarrow \overline{L_1 \cap L_2}$$

Answer:

$$L_1 \cap L_2 = \Sigma 1 \Sigma$$

So,

$$(L_1 \cap L_2)' = \Sigma^* - \Sigma 1 \Sigma$$

This means **all binary strings except the length-3 strings whose second symbol is 1.**

Therefore:

$$\epsilon + \Sigma + \Sigma^2 + \Sigma 0 \Sigma + \Sigma^4 \Sigma^*$$

The complement is represented by these cases:

- $\epsilon + \Sigma + \Sigma^2$: lengths below 3
- $\Sigma 0 \Sigma$: length 3, but second symbol is 0
- $\Sigma^4 \Sigma^*$: length at least 4

So the excluded strings are exactly:

$$\{010, 011, 110, 111\}.$$